# Samira Huber

## PhD Candidate | Embodied AI · Robot Learning · World Models

PhD candidate in **embodied AI** and **robot learning**, researching world models and foundation models. First-author publications at **CoRL** and **IROS**, with hands-on experience developing and deploying learning-based systems on physical robots using ROS2 and simulation.

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## PROFESSIONAL EXPERIENCE

**PhD Candidate: University of Kiel** Sep 2025 – Present
Embodied AI & Robotics | World models and robot learning for real-world robotic autonomy | Developing and evaluating learning-based systems in simulation and on physical robots | First-author publications at CoRL 2026 and IROS 2026.

**CTO & Co-Founder: EcoCheck** Nov 2021 – Present
AI Systems & Entrepreneurship | Spearheaded AI-powered $CO_2$ accounting systems

**Freelancer: Software Development & AI** May 2022 – Aug 2025
Delivered custom AI-driven software solutions | Provided specialized IT consulting for diverse clients

**Working Student: Tesla Inc., Brandenburg** Oct 2021 – Mar 2022
Developed scalable React-based web applications | Full-Stack Development | For production monitoring

**Working Student: Abusix Inc., Karlsruhe** Oct 2020 – Oct 2021
Full-Stack Development | Built core product features | Implemented frontend and backend solutions using React

**Working Student: Intelligent Media Systems AG, Karlsruhe** May 2018 – Sep 2019
ML Systems & APIs | Trained scalable recommender systems | Developed REST APIs and refactored platforms in Vue.js

## SELECTED RESEARCH

**COOL: Curiosity-Driven Object Ownership Learning for Personalized Robotic Assistance** CoRL 2026
First Author | Embodied AI | Personalized Robot Assistance | Real-World Robotics | ROS2

**HUMEMBR: Learning Human Routines for Predictive Embodied Navigation** IROS 2026
First Author | Embodied AI | Human Behavior Modeling | Real-World Navigation

**Hydra: A Navigation World Action Model with Discrete Latent Planning and Continuous Flow-Matching Execution** Submitted
Co-Author | World Models | Robot Navigation | Real-World Robotics | ROS

**PUSHER: Learning Vision-based Policies for Non-Prehensile Object Manipulation with Quadruped Robots** Submitted
Co-Author | Reinforcement Learning | Isaac Sim

**Evaluating Early Multimodal Fusion in Discrete Latent World Models for Robot Navigation** Submitted
Co-Author | World Models

**What Actually Matters? A Paired Evaluation of Inference Settings for $\pi_{0.5}$ on BEHAVIOR-1K** Preprint
First Author | Vision-Language-Action Model | Manipulation + Navigation

**Fire-Aware Navigation** Work in Progress
First Author | Reinforcement Learning | Isaac Sim

## EDUCATION

**University of Lübeck** Oct 2020 – Dec 2022
*M.Sc. Entrepreneurship in Digital Technologies*, Grade 1.3
Thesis: Identification of barriers to sustainable shopping and methods to address them.

**Baden-Württemberg Cooperative State University** May 2018 – Sep 2020
*B.Sc. Information Technology*, Grade 1.8
Thesis: Implementation of a recommender system for web-based training platforms.

**Baden-Württemberg Cooperative State University** Aug 2017 – Apr 2018
*B.Eng. Mechatronics* (25 ECTS completed)